

Examiner's Report

PhD thesis of Lu Niu

Exploring Quantum Properties of Light-Element Xenon: DFT Studies of the Electronic Structure and Optical Response for Boron and Beryllium Allotropes

This thesis presents a substantial programme of first-principles research on light-element two-dimensional materials. It studies graphene-based boron heterostructures, boron allotropes, and pristine or hydrogen-functionalised beryllene, with attention to electronic structure, optical response, magnetism, superconductivity, stability, and topology. The topic is timely and the combination of structural and property calculations is potentially valuable for the design of new low-dimensional materials.

The thesis contains original computational research. The candidate has published 2 papers (Chapters 4 and 5), and is preparing another paper (Chapter 6). The main contributions appear to be the study of graphene-based boron heterostructures, the predicted magnetic and superconducting properties of o -B₁₄, and the proposed cubic trilayer beryllene structure. The candidate's individual contribution to the multi-author chapters should be stated more explicitly, and the revised thesis should distinguish genuinely new predictions from the characterisation of known structures.

The research scope and publication record are appropriate for a PhD, and the presentation is generally professional. The thesis-level research position is nevertheless not sufficiently clear: the early literature review does not lead to a sharp research gap or central motivation. Chapters 2 and 3 together are also disproportionately long, at approximately 100 pages, while still not giving the reader a concise roadmap of the computational methods (geometry optimisation, phonon, magnetism, AIMD, superconductivity, topological property) used later. Substantial scientific and editorial corrections are therefore required. The rest chapters look like hopping between different materials and there is a lack of relationship between them. More detailed comments are given in the context below.

Chapter 1 provides useful background but reads more as a broad survey than as a focused argument for the thesis. It should explain why graphene-based boron systems, boron phases, and beryllene belong to one thesis, what unresolved question each chapter addresses, and how the studies collectively support one conclusion. The literature should be used critically to separate established experimental systems from theoretical proposals and to identify the research gap. The motivation for calculating optical properties is also not sufficiently clear: absorption, refractive index, reflectivity, and loss spectra are often presented as a standard set of outputs rather than as observables answering defined scientific questions. The introduction should also briefly explain why topology is important and what physical consequence is expected from a non-trivial Z_2 invariant, before introducing the parity calculation.

Chapter 2 and 3 are very long and contain considerable formal background, but they do not provide a sufficiently concise and practical explanation of the computational methods used in the thesis. The reader more needs a brief, method-oriented account of geometry optimisation and convergence criteria, Brillouin-zone sampling and the choice of k -points, phonon calculations, AIMD, spin-polarised calculations and magnetic ordering, electron-phonon calculations and superconductivity, and spin-orbit coupling and topological analysis. These methods should be connected explicitly to the scientific questions in the later chapters. These chapters should be shortened and reorganised around the methods actually used.

Chapter 4 is a useful but relatively incremental computational study. It addresses the lack of systematic knowledge about the electronic and optical properties of graphene-boron-based heterostructures. Its main contribution is mapping structure-property relationships and suggest-

ing optical/plasmonic tunability, rather than solving a major practical problem. The captions of Figures 4.29–4.36 are too short to be self-contained and should identify the panels, tensor components, line styles, PBE–HSE06 comparison, polarisation directions, units, and principal feature.

Chapter 5 is stronger and more scientifically interesting. It asks how dimensionality and hydrogen functionalisation modify the properties of a complex boron phase, including magnetism, optical response, and superconductivity. This is a meaningful materials-design problem. However, the work remains predictive: the stability, superconductivity, and plasmon claims require careful qualification and are not experimentally demonstrated. This chapter is scientifically substantial and includes published work on *o*–B₁₄, its magnetic states, optical response, and hydrogen-terminated bilayer superconductivity. Cohesive energies, phonons, and a very short 5.5 ps AIMD trajectory establish energetic preference relative to selected references, dynamical stability, and short-time thermal persistence, but do not by themselves establish thermodynamic stability or synthesizability. The small AFM–FM energy difference and inconsistent statements about negative phonon frequencies should be clarified. The monolayer is described as an AFM semiconductor in one place and as metallic in the optical discussion, so the electronic and magnetic state used for the optical calculation and its intraband treatment must be stated. The PBE–HSE06 difference should be explained: HSE06 can open a gap by reducing self-interaction and delocalisation errors, but it remains an approximation. Finally, the superconductivity workflow needs a concise reproducible explanation of how density-functional perturbation theory, Wannier interpolation, and EPW produce the phonons, electron–phonon matrix elements, $\alpha^2 F(\omega)$, λ , $\omega_{\log}$, and the Allen–Dynes and anisotropic Migdal–Eliashberg values. The distinction between phonon stability and finite-temperature AIMD should be made clear.

Chapter 6 is potentially the most novel. It proposes a new cubic trilayer beryllene structure and explores whether elemental beryllium can support tunable electronic, optical, and topological properties. The main weaknesses are that the method used to generate the new structure is unclear, and the Z_2 claim needs justification because the phase is described as metallic. Details of how the new structure was constructed should be described. Stability and optical quantities should be reported with appropriate two-dimensional normalisation and with a clear distinction between dynamical stability, energetic preference, and experimental synthesizability.

Overall, the thesis contains substantial original research and is capable of reaching a professional doctoral standard, but the research gap, computational-methods overview, and scope of several scientific claims need to be clarified before acceptance.